

Uncertainty Quantification and Calibration in Template-Based Retrosynthesis Models: A Cross-Architecture Study of LocalRetro and MHNreact

Yamen Fargaly*¹

¹Princeton University

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Abstract

Template-based single-step retrosynthesis models output a softmax distribution over a template vocabulary, and their top score is routinely read as a confidence. We ask whether that built-in signal reliably separates correct from incorrect predictions, and whether the separation survives distribution shift toward rare and novel templates. Across two architecturally distinct models on USPTO-50K ($n = 5,007$), LocalRetro (a graph neural network) and MHNreact (a Modern Hopfield retrieval network), the answer is no, and the failure takes two distinct forms. LocalRetro is well calibrated in aggregate but collapses sharply under template rarity; MHNreact is more uniformly miscalibrated. The standard perturbation remedies, MC dropout and deep ensembling, do not rescue the rare tail in either model, and in MHNreact both signals are inverted rather than merely uninformative, a boundary-robust and seed-replicated result. Per-tier (Mondrian) conformal selective prediction is the one method that restores error control. Finally, a held-out-reaction-class experiment confirms the collapse under genuine novelty: with two entire reaction classes removed from training, top-1 accuracy falls to 0.00–0.25% while the model remains confident (out-of-distribution ECE up to 0.39), the deployment failure mode that matters most.

Keywords

uncertainty quantification; calibration; retrosynthesis prediction; template-based models; conformal prediction; deep learning; distribution shift; graph neural networks

*Corresponding author. Email: yf0550@princeton.edu. ORCID: <https://orcid.org/0009-0000-5289-6265>

1 Introduction

This paper presents a cross-architecture uncertainty quantification (UQ) and calibration study of template-based retrosynthesis models. It combines an end-to-end analysis of LocalRetro, a graph neural network for single-step retrosynthesis prediction,¹ with a second, architecturally distinct model, MHNreact,² a Modern Hopfield Network operating on molecular fingerprints. LocalRetro combines local atom and bond reactivity scoring with template based reactant prediction; MHNreact instead encodes molecules as fingerprints and retrieves templates via Hopfield association. Both output a softmax distribution over a template vocabulary, which is what makes the calibration and OOD-frequency methodology transferable between them.

The question we set out to answer: does a template-based model’s built in confidence signal ($1 - \text{top_score}$) reliably separate correct from incorrect predictions, and does that separation survive under distribution shift toward rare, novel templates? The headline result is that it does not: **confidence calibration collapses under template rarity in both models**, with two distinct failure signatures (LocalRetro suffers a sharp distribution-shift collapse; MHNreact is more uniformly miscalibrated). The obvious perturbation-based fixes, MC dropout and deep ensembling, do not rescue the rare tail in either model, and in MHNreact both perturbation signals are not merely uninformative but *inverted*, a boundary-robust, seed-replicated finding. What does work is per-tier (Mondrian) conformal selective prediction, an architecture-agnostic fix that correctly abstains where no threshold can deliver a reliable guarantee.

Every result below comes from a single trained checkpoint per model, with template extraction, preprocessing, and all downstream analyses run against that fixed checkpoint (see Section 3). Figures and tables presenting the full result in both architectures are in the main text; single-architecture detail, distribution plots, and ROC curves that support (without adding to) the main-text AUROC numbers are collected in the Supplementary Information at the end of this document.

Contributions

1. **First calibration-under-shift study for template-based retrosynthesis.** We characterize how the built-in confidence signal of single-step template models behaves under template-frequency shift, across two architectures (GNN and Modern Hopfield retrieval) on a common benchmark.
2. **Two failure signatures, not one.** We show the miscalibration is architecture-dependent: a sharp distribution-shift collapse in LocalRetro versus uniform miscalibration in MHNreact, distinguishing model-specific from general behavior.
3. **A negative result on perturbation UQ.** MC dropout and deep ensembling do not rescue the rare tail in either model, and are inverted in MHNreact. We harden this with a boundary sweep, a second independent ensemble, and a named benign mechanism (multi-valid-template ties), so it is reported as a robust finding rather than an artifact.
4. **A working mitigation and a genuine-novelty confirmation.** Mondrian conformal selective prediction restores per-tier error control, and a held-out-reaction-class experiment reproduces the confident-but-wrong collapse under true out-of-distribution novelty (0% accuracy at substantial confidence), moving the result from suggestive diagnostic to demonstrated phenomenon.

Key Findings

- Baseline confidence ($1 - \text{top_score}$) separates correct from incorrect predictions with AUROC = 0.710 (LocalRetro) and 0.738 (MHNreact), a real and statistically significant effect in both models (LocalRetro: Spearman $\rho = 0.364$, $p < 10^{-150}$).
- **The most consequential finding: calibration degrades with template rarity in both models.** Using rank-based vocabulary-matched tiers (exact same n per tier for both models), both show highest calibration error in the rare tail, though with different profiles: the rare/high ratio is $3.2\times$ vs $2.9\times$, but the rare/mid ratio is $7.3\times$ vs $1.8\times$, reflecting genuinely different failure shapes. LocalRetro collapses sharply under novelty (debaised ECE 0.024 to 0.291); MHNreact (retrained with verified early stopping, Section 3.1) shows degradation concentrated in the tail (0.038 to 0.107 on absolute-count tiers), with

a clean monotonic gradient on the rank-based tiers. In LocalRetro’s very-rare tier ($n=100$), confidence discrimination is statistically indistinguishable from random (AUROC 0.464, 95% CI [0.248, 0.706]).

- MC dropout (20 stochastic passes, 4 dropout layers) does not improve on baseline for LocalRetro. Prediction variance across passes (`std_top_score`) is a weak signal on its own for LocalRetro (AUROC = 0.545), and no combination we tried beat the simple softmax confidence. MHNreact’s MC dropout variance behaves very differently: it is not merely weak but inverted (AUROC = 0.330, overall): higher dropout variance predicts a *correct* prediction more often than an incorrect one, and this holds up under a boundary-robustness sweep (Section 7).
- A 4-seed deep ensemble improves overall discrimination for LocalRetro (paired DeLong test, $\text{diff} = 0.017$, $p = 3.6 \times 10^{-8}$) but not for MHNreact ($\text{diff} = 0.001$, $p = 0.868$). Worse, MHNreact’s ensemble disagreement is inverted in the very-rare tier (AUROC = 0.427, Bonferroni CI [0.394, 0.461]), the same direction of surprise as the MC dropout inversion, and this replicates on a second, completely independent 4-seed ensemble (AUROC = 0.4277) and across every very-rare-tier boundary tested.
- Mondrian (per-tier) conformal prediction is the proposed fix, and is architecture-agnostic. A global threshold catastrophically fails the very-rare tier (keeps 9% of predictions with a 100% error rate). A per-tier Mondrian threshold abstains almost entirely on the very-rare tier and lands close to the error target elsewhere, for both models, on a single calibration split. A follow-up Hoeffding-UCB-corrected threshold, run independently on both architectures, shows the naive version has an optimism bias in every tier except very-common for both models; the corrected version is honest but often equivalent to abstaining until more calibration data is available.
- **A genuine held-out-reaction-class experiment (Section 6) confirms the frequency-proxy collapse was, if anything, an understatement.** On truly novel reaction classes, LocalRetro scores 0/909 and 0/811 correct across two independent classes (architecturally impossible, since the held-out templates do not exist in its retrained vocabulary) and MHNreact scores 1/909 and 2/811 (astronomically unlikely but not architecturally forbidden, since its vocabulary is fixed upfront). Both models remain confidently wrong (OOD ECE 0.376/0.390 for LocalRetro, 0.268/0.257 for MHNreact), the practical failure mode that matters most for deployment.
- **Retiering by the predicted, rather than true, template frequency (the deployment-realistic scenario) changes the picture for both models.** LocalRetro’s true-frequency gradient (11% to 57% accuracy) partially inverts under predicted-frequency tiering (33% to 53%, peaking in the common tier); MHNreact’s gradient (28% to 67%) compresses to a much flatter 39% to 50% but stays monotonic. The contrast between the two tiering schemes mirrors each model’s own established failure signature.

2 Related Work

Calibration and uncertainty in neural networks. Modern classifiers are frequently miscalibrated,³ and the standard remedies are temperature scaling, deep ensembles,⁴ and MC-dropout Bayesian approximation.⁵ Calibration is usually measured with expected calibration error.⁶ Our work applies this toolkit to a domain where it has not been characterized and reports where the perturbation-based members of it fail.

Conformal and selective prediction. Conformal prediction gives distribution-free coverage guarantees,^{7,8} and Mondrian/class-conditional variants provide per-group control.⁹ We use Mondrian conformal selective prediction as the mitigation that works when perturbation UQ does not, and analyze where finite rare-tier sample sizes bound it.

Uncertainty in molecular and reaction models. UQ has been benchmarked for molecular property prediction (Hirschfeld et al.;¹⁰ Heid et al.¹¹) and for calibration and generalizability of probabilistic models on small chemical datasets (Tom et al.¹²), and for reaction-level prediction such as yield forecasting (Kwon et al.¹³). The closest prior calibration work on reaction outcomes is Schwaller et al.,¹⁴ which calibrates a template-free transformer for *forward* reaction prediction. To our knowledge no prior work studies calibration of template-based *single-step retrosynthesis* models under template-frequency distribution shift, nor compares this behavior across architecturally distinct models, which is the gap this study fills.

3 Methodology

Sections 3.1–3.3 describe the LocalRetro and MHNreact training pipelines; Sections 3.4–3.6 describe the shared OOD-frequency proxy, calibration metrics, and UQ signal protocols applied identically to both models.

3.1 Model and Data Pipeline

LocalRetro was trained from scratch on the standard USPTO-50K dataset (Schneider/GLN split: 40,009 train, 5,002 validation, 5,008 test reactions), using the official preprocessing pipeline (`Extract_from_train_data.py`, then `Run_preprocessing.py`, then `Train.py`). Template extraction and training used the same fixed checkpoint throughout all downstream analyses. Template extraction yielded 124 atom level templates and 547 bond level templates. Training used early stopping (patience = 5) and converged at epoch 19 of 50, with validation loss 0.0324 and test loss 0.0319. Top 1 accuracy on the full standard test set came out to 53.11% (2,659 of 5,007 correct), consistent with LocalRetro’s published stereo unaware top 1 accuracy range.

MHNreact² was trained from scratch on the same USPTO-50K standard split (Morgan fingerprint size 4096, RDKit template fingerprints, Hopfield association dimension 512) using the official training script. Training used early stopping on validation loss (patience 5), converging at epoch 6 with validation loss 3.193; this stopping point was independently verified by rerunning with patience 10, which found no improvement over the remaining 10 epochs and converged to the identical checkpoint. Top-1 accuracy on the test set: 46.7%. Correctness labels were obtained by applying each predicted template to the product SMILES via `rdchiral` and comparing the resulting reactants to the ground truth using RDKit canonical SMILES with fragment-order-independent comparison. Template frequency was computed from MHNreact’s own training data using its 11,800-template vocabulary, and the same four frequency tiers were applied.

3.2 Baseline UQ Calibration

For each test molecule, $UQ = 1 - \text{top_score}$, where `top_score` is the softmax probability assigned to the model’s top 1 predicted template (for LocalRetro, the (template, site) edit). Correctness (error = 0 or 1) was determined by canonicalizing the predicted top 1 reactant SMILES (RDKit, fragment order independent) and comparing it against the ground truth reactants.

3.3 MC Dropout

For LocalRetro, all `nn.Dropout` submodules (4 found in the trained model) were turned back on at inference while the rest of the model stayed in eval mode. For MHNreact, its `nn.Dropout` submodules (mol and template encoders, dropout $p = 0.2$) were reactivated the same way, verified against the model’s actual source rather than assumed. In both cases we ran 20 stochastic forward passes per molecule, producing `mean_top_score` (averaged prediction confidence), `std_top_score` (variance across passes), `agreement_rate` (fraction of passes agreeing with the consensus top 1 prediction), and `predictive_entropy`.

3.4 Out of Distribution Proxy: Template Training Frequency

We did not run a full held out reaction class retrain, since no reaction class labels were available for this dataset instance. Instead, each test reaction’s true template’s frequency in the training set served as a continuous proxy for how in distribution or out of distribution that reaction is. Templates seen thousands of times in training are about as in distribution as it gets; templates seen once or twice are functionally rare and OOD like without requiring any retraining. Test molecules were bucketed into four absolute-count frequency tiers (1 to 5, 6 to 20, 21 to 100, 100+ training occurrences) for within-model analysis. Because LocalRetro (671 templates) and MHNreact (11,800 templates) have very different vocabulary sizes, absolute-count tiers assign very different fractions of each test set to the same nominal tier, so cross-model comparison instead uses **rank-based** quantile cutting: each model’s test set is sorted by its own template’s training frequency and cut by rank position, forcing each tier to hold exactly the same number of molecules (n=500, 1,152, 1,652, 1,703) in both models regardless of vocabulary size or value ties.

3.5 Calibration Metrics: Reliability Diagrams, ECE, and Conformal Prediction

All AUROC values reported measure *discrimination* (does the signal rank correct predictions above incorrect ones?), not *calibration* (does stated confidence match observed accuracy?). We supplement discrimination results with: (1) reliability diagrams and ECE/MCE, both overall and per frequency tier; (2) equal-mass (quantile) binning alongside standard equal-width binning, since fixed-bin ECE has a positive small-sample bias at small n that would artificially inflate the rare-tier vs. very-common-tier comparison; and (3) a simulated bias floor (500 replicates of perfectly-calibrated synthetic data at each tier’s actual confidence distribution and n) to report a debiased ECE. All AUROC values carry bootstrap 95% CIs (2,000 resamples). For rare-tier multi-signal comparisons, we additionally report Bonferroni-corrected CIs across the signals compared within that tier.

3.6 Mondrian Conformal Selective Prediction

We implement Mondrian (frequency-stratified) conformal selective prediction as a proposed fix. Standard (marginal) conformal sets a single score threshold targeting a global error rate; Mondrian sets a separate threshold per frequency tier, restoring the coverage guarantee within each tier independently.

Tiering. Reactions are partitioned into four tiers by the training-set occurrence count of the reaction’s template: $T_1 = 1\text{--}5$ (very rare), $T_2 = 6\text{--}20$ (rare), $T_3 = 21\text{--}100$ (common), $T_4 = 100+$ (very common). For the Mondrian selective-prediction results in this section, the tier is keyed on the *true* template’s frequency; Section 5 additionally reports a direct comparison against *predicted*-template-frequency tiering, the deployment-realistic scheme, since the true template is unknown at inference.

Nonconformity score. For each molecule i , a logistic model is fit on the calibration split, predicting error from two UQ features:

$$s_i = \sigma(b_0 + b_1(1 - p_i) + b_2 \text{std}_i) \quad (1)$$

where $p_i = \text{top_score}$, $\text{std}_i = \text{MC-dropout standard deviation across passes}$, and s_i is the predicted probability the prediction is wrong. Higher s_i means less trustworthy.

Per-tier threshold selection (the Mondrian step). For each tier T_k independently, using only calibration points in that tier, the most permissive threshold τ_k is chosen whose empirical error rate among kept predictions stays at or below the target α ($= 0.20$):

$$\tau_k = \max \{ \tau : \text{err}_k(\tau) \leq \alpha \}, \quad \text{err}_k(\tau) = \frac{\#\{i \in T_k^{\text{cal}} : s_i \leq \tau, \text{ wrong}\}}{\#\{i \in T_k^{\text{cal}} : s_i \leq \tau\}} \quad (2)$$

At test time, prediction i in tier k is kept if $s_i \leq \tau_k$; otherwise the model abstains. Maximizing τ_k maximizes coverage subject to the error guarantee. If no τ satisfies the constraint, the tier abstains entirely, which is what happens to the very-rare tier in both models.

Hoeffding-UCB correction. Because τ_k is chosen by searching many thresholds and keeping the best-looking one, the empirical error in Equation 2 is optimistic. The corrected variant instead requires the upper confidence bound on the error to clear the target, at confidence $1 - \delta$ ($\delta = 0.1$):

$$\text{err}_k(\tau) + \sqrt{\frac{\ln(1/\delta)}{2 n_k(\tau)}} \leq \alpha \quad (3)$$

where $n_k(\tau)$ is the number of calibration points kept at threshold τ in tier k . This is the honesty check that determines whether the rare and common tiers can support a valid guarantee at all, rather than an optimistic one (Section 8).

Full results, including per-tier coverage and error under both the naive and Hoeffding-UCB-corrected thresholds, are in Section 8.

4 Baseline Confidence: A Moderate but Real Discriminator (Both Models)

The softmax derived confidence signal shows a real, statistically significant ability to separate correct from incorrect predictions in both models. LocalRetro’s overall ECE (0.051) is noticeably lower than MHNreact’s (0.087), though both mask substantial per-tier variation, which is the subject of Section 5.

Metric	LocalRetro (GNN)	MHNreact (Hopfield)
Top 1 Accuracy	53.11% (2,659 / 5,007)	46.69% (2,338 / 5,007)
AUROC (1 – top_score vs. error)	0.7104 [0.696, 0.724]	0.7376 [0.724, 0.752]
Spearman ρ	0.3637 ($p = 2.04 \times 10^{-156}$)	0.4107 ($p = 4.48 \times 10^{-203}$)
ECE (overall, 10 bins)	0.051	0.087
MCE (overall)	0.214	0.154
Mean UQ correct predictions	0.367	0.381
Mean UQ incorrect predictions	0.534	0.627

Table 1: Baseline UQ performance, both models (MHNreact retrained with verified early stopping, Section 3.1). Both signals discriminate correct from incorrect predictions; MHNreact’s higher Spearman ρ but higher overall ECE reflects the two-failure-signatures pattern developed in Section 5: MHNreact separates classes slightly better in rank but is less well calibrated in absolute probability terms.

The overall ECE in Table 1 is an average, and the average is dominated by easy cases. For LocalRetro, 85% of the test set (4,245 of 5,007 reactions) falls in the very-common tier, where debiased ECE is 0.024; the overall figure of 0.051 is low largely because those 4,245 easy cases outnumber the 276 rare-tail cases (6%) where calibration is 8–12 $\times$ worse (Section 5). Table 1 is not wrong, but taken alone it is misleading: it is the single-number summary the rest of this paper exists to break apart. This is not an isolated glitch confined to the very-rare bin: LocalRetro’s debiased ECE rises smoothly and monotonically across all four tiers (0.024, 0.113, 0.200, 0.291 from very-common to very-rare, Section 5), and the held-out-class experiment (Section 6) shows that under genuine novelty, rather than mere rarity, the floor gives way entirely: 0% accuracy at roughly 38% confidence. Figure 1 shows the same models’ calibration once the aggregate is disaggregated by tier, immediately beside the aggregate view rather than several pages later, so the mask and the reality are visible together.

This is setup, not the contribution: both models show a moderate, real, but unremarkable discrimination signal in-distribution once disaggregated correctly. The interesting behavior, where that signal survives and where it fails, is the subject of the rest of this paper. Supporting figures (mean UQ by correctness) are in Supplementary Information Section S1.

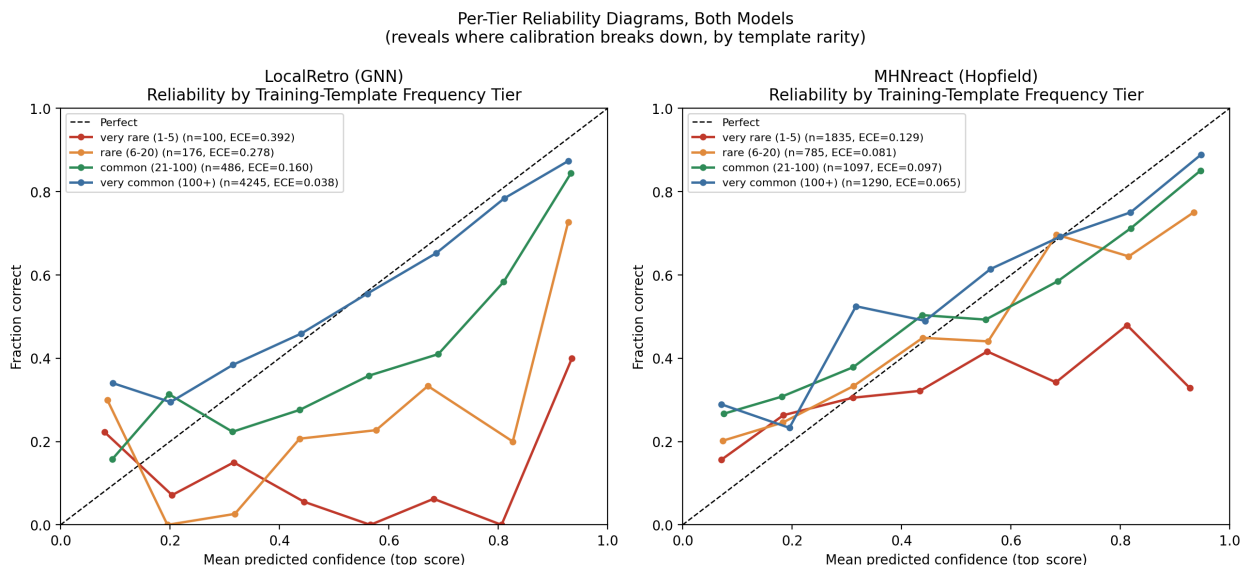


Figure 1: Reliability by training-template frequency tier, both models, shown alongside Table 1’s aggregate numbers to make immediately visible what the aggregate view hides (side-by-side panels, since each model’s tier boundaries are defined over a different-sized vocabulary: 671 templates for LocalRetro vs. 11,800 for MHNreact, so the tiers are not the same population of reactions and are not meaningfully overlaid on one set of axes). *Left (LocalRetro)*: tiers fan out sharply: very-common (blue) tracks the diagonal closely while very-rare (red, n=100) is noisy and far below it, visually matching the twelve-fold debiased ECE spread (0.024 to 0.291) despite an overall ECE of only 0.051. *Right (MHNreact)*: tiers stay comparatively clustered together across the whole confidence range, consistent with the uniformly-mildly-miscalibrated signature.

5 Calibration Collapses Under Template Rarity (Both Models)

This is the central finding of the paper. Both expected directional correlations hold in both models: rarer training templates correlate with more errors and with lower confidence, but the magnitudes are starkly mismatched between how fast accuracy collapses and how much confidence actually adjusts. The two models collapse in visibly different ways.

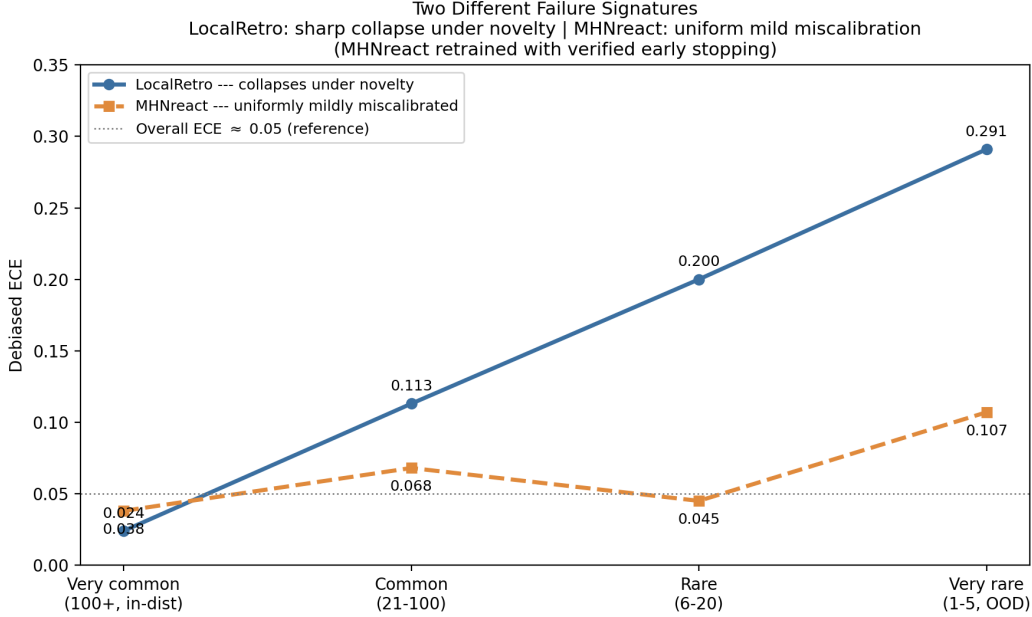


Figure 2: Two different failure signatures. LocalRetro is well-calibrated in-distribution (debiased ECE 0.024 at very common tier) and collapses sharply under novelty (0.291 at very rare). MHNreact (retrained with verified early stopping) is mildly but uniformly miscalibrated across all tiers (0.038–0.107), with no sharp collapse. Both patterns are well above the ideal (ECE = 0); they represent distinct miscalibration regimes, not one model being simply better than the other.

The contrast:

- **LocalRetro:** well-calibrated in-distribution (debiased ECE = 0.024 for very common templates), collapses sharply under novelty (0.291 for very rare). A model that is trustworthy in familiar territory and unreliable under distribution shift. Across the four absolute-count frequency tiers, accuracy collapses from 57.3% (templates seen 100+ times) to just 10.0% (templates seen 1 to 5 times), a 47.3 point drop, while mean UQ rises by only 0.110 over the same range. In the very-rare tier (n=100), confidence discrimination is statistically indistinguishable from random (AUROC 0.464, 95% CI [0.248, 0.706]).
- **MHNreact:** mildly but uniformly miscalibrated (debiased ECE ranging 0.038–0.107, no sharp collapse; retrained with verified early stopping, Section 3.1). A model with a more diffuse, architecture-wide calibration offset. MHNreact’s rare-tier debiased ECE (0.107) remains lower than LocalRetro’s *common-tier* ECE (0.113), though the margin has narrowed from the pre-retraining comparison (0.096 vs. 0.113). Unlike LocalRetro, no MHNreact tier’s confidence-discrimination CI comes close to spanning random performance; MHNreact retains real discrimination even in its very-rare tier (AUROC 0.634 [0.608, 0.661]).

Both models show their worst calibration in the rare tail; the shared degradation direction is the cross-architecture finding. The specific failure mode differs: LocalRetro suffers a distribution-shift collapse; MHNreact suffers a more uniform overconfidence. Mondrian conformal selective prediction (Section 8) addresses both, since it is agnostic to the shape of the ECE profile, requiring only the softmax confidence score and the template frequency label. Per-tier reliability diagrams for both models are in Supplementary Information

Rank-Based Vocabulary-Matched Confirmation

The two models use very different template vocabularies (LocalRetro: 671 templates; MHNreact: 11,800 templates), so absolute-count frequency tiers assign different fractions of the test set to each tier: LocalRetro’s very-rare tier (≤ 5 training occurrences) contains 100 molecules (2% of test set); MHNreact’s contains 1,835 molecules (36.6%). These are not the same population of reactions viewed through two models; they are different slicings, and a naive value-based quantile cut also fails to produce matched tiers because thousands of MHNreact templates share frequency=1. Rank-based quantile cutting (sort each model’s test set by its template’s training frequency, then cut by rank position) forces each tier to hold exactly the target fraction regardless of ties, giving four tiers of $n=500, 1,152, 1,652, 1,703$, identical for both models.

Tier (by rank)	LocalRetro (GNN)				MHNreact (Hopfield)			
	n	Acc	ECE deb	AUROC	n	Acc	ECE deb	AUROC
Rare (bot 10%)	500	0.252	0.197	0.642 [0.579, 0.702]	500	0.148	0.134	0.520 [0.456, 0.588]
Low (10–33%)	1,152	0.373	0.135	0.712 [0.680, 0.743]	1,152	0.309	0.093	0.644 [0.611, 0.677]
Mid (33–66%)	1,652	0.583	0.027	0.706 [0.682, 0.732]	1,652	0.490	0.076	0.708 [0.681, 0.733]
High (top 33%)	1,703	0.669	0.061	0.699 [0.671, 0.726]	1,703	0.645	0.047	0.741 [0.717, 0.764]
rare/high ratio			3.2×				2.9×	
rare/mid ratio			7.3×				1.8×	

Table 2: Rank-based vocabulary-matched tier comparison (MHNreact retrained with verified early stopping; Section 3.1). Each tier holds exactly the same number of molecules for both models. The rare/high ratios ($3.2\times$ vs $2.9\times$) are similar, but the rare/mid ratios ($7.3\times$ vs $1.8\times$) differ substantially, reflecting different ECE profiles across the frequency spectrum. All AUROC values carry 95% bootstrap CIs (2,000 resamples). LocalRetro’s debiased ECE series is not strictly monotonic (the high tier exceeds the mid tier); MHNreact’s is now strictly monotonic following retraining.

MHNreact was retrained from scratch with early stopping on validation loss (patience 5, converged at epoch 6, validation loss 3.193), achieving 46.7% top-1 accuracy, essentially unchanged from an earlier, methodologically flawed fixed-10-epoch run (46.6%) that had saved its final epoch rather than its best validation checkpoint. This retraining, and the confirmation that it did not materially change any result reported here, is detailed in Section 10. The rank-matched ECE/AUROC comparison is shown visually in Supplementary Information Section S2.

A Single Global Temperature Cannot Fix a Tier-Dependent Problem

Before testing perturbation-based fixes (Section 7), we test the simplest and most standard recalibration remedy: temperature scaling,³ a single scalar T fit to minimize calibration error, applied post-hoc to the model’s confidence. Since only the top-1 softmax probability is available (not the full logit vector), we use the standard one-vs-rest approximation: T is fit on a held-out calibration half-split to minimize negative log-likelihood of $p^{1/T}/(p^{1/T} + (1-p)^{1/T})$ against correctness, then applied to the test half.

The overall ECE improvement looks substantial for both models, but the per-tier breakdown (Table 3) shows why that is misleading (shown visually in Supplementary Information Section S4). For LocalRetro, essentially all of the overall gain comes from the very-common tier (85% of the data); the very-rare and rare tiers barely move ($0.408\rightarrow 0.404$, $0.299\rightarrow 0.296$), and the common tier does not move at all ($0.158\rightarrow 0.158$). A single scalar cannot repair LocalRetro’s sharp, tier-concentrated collapse, because it applies the same correction everywhere regardless of tier. For MHNreact, the improvement is more evenly spread across tiers, including a real (if partial) gain in the rare tiers ($0.128\rightarrow 0.101$, $0.117\rightarrow 0.057$), consistent with MHNreact’s own uniformly-miscalibrated signature (Figure 2) rather than a sharp collapse: a global correction works better when the underlying miscalibration is itself more global. Either way, neither model’s rare tail is fixed by temperature scaling to the degree the overall number suggests, which is exactly the failure mode Mondrian conformal (Section 8) is designed around: a per-tier problem needs a per-tier fix, not a global one.

Tier	n	ECE (raw)	ECE (T-scaled)
<i>LocalRetro</i> (fitted $T = 1.513$)			
Very rare (1–5)	46	0.408	0.404
Rare (6–20)	84	0.299	0.296
Common (21–100)	243	0.158	0.158
Very common (100+)	2,131	0.040	0.031
Overall	2,504	0.047	0.028
<i>MHNreact</i> (fitted $T = 1.744$)			
Very rare (1–5)	917	0.128	0.101
Rare (6–20)	403	0.117	0.057
Common (21–100)	548	0.104	0.034
Very common (100+)	636	0.060	0.037
Overall	2,504	0.099	0.035

Table 3: Temperature scaling, evaluated per tier on a held-out calibration/test split (independent of the main-text tables’ binning scheme, so these ECE values are not directly comparable to Tables 1/2 in absolute terms; the comparison that matters is raw vs. T-scaled within this table). Both fitted temperatures exceed 1, consistent with both models being overconfident on average.

Single-architecture detail (LocalRetro’s and MHNreact’s own absolute-count-tier calibration gap, AUROC-by-tier, and fine-grained 10-quantile-bin figures, plus the full absolute-count tier table) is in Supplementary Information Section S3, since the rank-based comparison above is the primary, confound-free cross-model result.

True-Template vs. Predicted-Template Tiering: The Deployment-Realistic Comparison

Every result above tiers each reaction by its *true* template’s training frequency, an honest analytical choice but not what a deployed tool actually has access to: at inference time, the correct template is unknown, and only the model’s *predicted* template is available. We test directly whether retiering by predicted-template frequency changes the picture, using each model’s own top-1 predicted template ID, looked up against the same training-frequency dictionary used throughout this paper.

LocalRetro’s checkpoint caveat. The original checkpoint underlying every other LocalRetro result in this paper could not be recovered for this analysis (Section 10); this comparison uses an architecturally and procedurally identical ensemble checkpoint (seed 2) instead, since LocalRetro’s training code sets no explicit random seed and is therefore not exactly reproducible even by rerunning the identical procedure. LocalRetro’s raw output additionally reserves one template index per edit type as a “no resolvable template” class; the 1,046 test molecules (20.9%) landing in that class are excluded from the tiering comparison below, leaving $n = 3,961$. MHNreact’s original baseline checkpoint was used directly, with no substitution and no exclusions beyond 35 molecules (0.7%) whose predicted template had zero training occurrences, reported as a separate “novel” cell rather than folded into the tiering.

The reason for both patterns is the same underlying mechanism, and the tier-migration data confirms it directly. When a model errs, it disproportionately defaults to common, high-frequency templates rather than guessing another rare one: of LocalRetro’s 88 true-very-rare-tier reactions, only 12 are still predicted as very-rare, and 61 migrate to the predicted-very-common tier; of MHNreact’s 1,835 true-very-rare-tier reactions, only 981 remain in the predicted-very-rare tier. This dilutes the predicted-very-common tier with wrong guesses that were never genuinely easy cases, dragging its accuracy down relative to the true-tiering version (LocalRetro: 57.1% $\rightarrow$ 52.8%; MHNreact: 67.4% $\rightarrow$ 50.1%) and inflating the predicted-very-rare tier’s accuracy relative to the true-tiering version, since only the reactions the model is actually confident enough about to guess something genuinely rare survive in that predicted cell.

The two models still diverge in exactly the way their established failure signatures predict. LocalRetro’s sharp, tier-concentrated collapse (Section 5) means predicted retiering visibly disrupts

Tier	True-template tiering		Predicted-template tiering	
	n	Acc	n	Acc
<i>LocalRetro (seed2 checkpoint, $n = 3,961$ resolvable predictions)</i>				
Very rare (1–5)	88	0.114	12	0.333
Rare (6–20)	162	0.198	67	0.418
Common (21–100)	454	0.405	333	0.547
Very common (100+)	3,257	0.571	3,549	0.528
<i>MHNreact (original checkpoint, $n = 5,007$)</i>				
Novel (freq=0)	–	–	35	0.686
Very rare (1–5)	1,835	0.284	981	0.393
Rare (6–20)	785	0.454	826	0.450
Common (21–100)	1,097	0.540	1,343	0.479
Very common (100+)	1,290	0.674	1,822	0.501

Table 4: True-template vs. predicted-template tiering, both models. True-tiering accuracy figures match those reported elsewhere in this paper exactly (a consistency check on the merge). Predicted tiering uses each model’s own top-1 predicted template, looked up against the same training-frequency dictionary. LocalRetro’s very-rare predicted-tier n (12) is small; treat that specific cell as indicative rather than precise.

its gradient, producing a non-monotonic dip at the top. MHNreact’s diffuse, uniform miscalibration means predicted retiering compresses the whole gradient without breaking its monotonic shape. The same underlying phenomenon, retiering by what a deployed model actually outputs rather than what it should have output, manifests differently in each architecture, consistent with the two-failure-signatures framing developed throughout this paper.

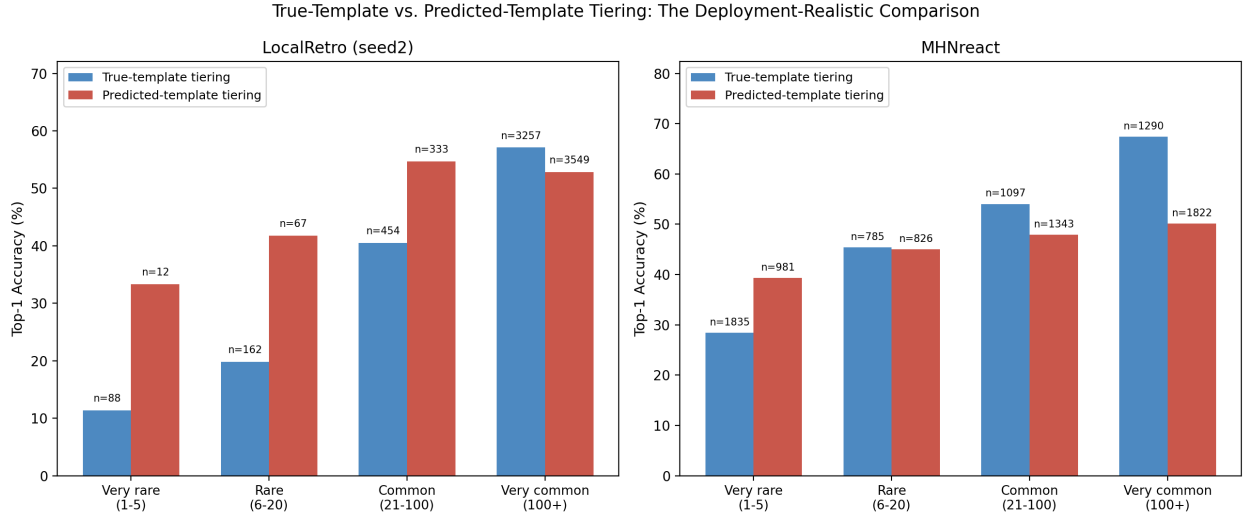


Figure 3: True-template vs. predicted-template tiering accuracy, both models. *Left (LocalRetro)*: the true-tiering gradient (11% to 57%) partially inverts under predicted tiering, which peaks in the common tier (54.7%) rather than very-common (52.8%). *Right (MHNreact)*: the true-tiering gradient (28% to 67%) compresses to a much flatter 39% to 50% under predicted tiering but remains monotonic. n per bar is annotated above each column.

6 True Distribution Shift: A Held-Out-Reaction-Class Experiment

Section 5 uses template training frequency as a continuous proxy for distributional novelty, an honest but indirect measure: a template seen once or twice is treated as functionally out-of-distribution without ever being genuinely absent from training. This section reports a stronger test: reaction classes held out entirely from training, so the templates involved are either structurally absent from the model’s vocabulary (LocalRetro) or present but never trained on (MHNreact, see below).

Design

Reaction classes follow the standard Schneider et al.¹⁵ 10-class taxonomy, the de facto labeling used across the USPTO-50K literature. Two classes were selected by a data-driven criterion, not chemical interest: template exclusivity. For each candidate class, we checked what fraction of its reactions use templates found *nowhere else* in the dataset, in both models’ own template vocabularies independently (LocalRetro’s 671-template space and MHNreact’s 11,800-template space are extracted separately and need not agree). Class 9, the largest candidate, turned out to be only 67.3% exclusive in LocalRetro’s vocabulary (its templates are shared with other classes over 30% of the time) and was rejected as a contaminated test. Class 4 (Heterocycle formation, n=909) and Class 8 (Oxidations, n=811 literature / 811 found) came back 99.8–100% exclusive in *both* models’ vocabularies independently, and were selected for that reason.

For each held-out class, all of its reactions (pooled across the original train/valid/test splits) are removed from training entirely and used as a genuine out-of-distribution (OOD) test set. A second, in-distribution (indist) test set, drawn from the original test split minus the held-out class, uses the same retrained model to confirm the retrain did not simply degrade the model overall.

The two architectures required different implementations because their template vocabularies are constructed differently, a distinction that matters for the results below rather than a methodological inconsistency. **LocalRetro** extracts its template vocabulary fresh from whatever training data it is given, so excluding a class’s reactions from training removes that class’s exclusive templates from the vocabulary entirely: a held-out template is structurally impossible to predict, not merely unlikely. **MHNreact**’s template vocabulary (11,800 templates) is fixed once, upfront, from the full dataset before any training run; excluding a class’s reactions from training leaves its templates present as valid output classes with random, untrained embeddings: a held-out template is astronomically unlikely to be predicted correctly, but not architecturally forbidden.

Results

The full per-class results are in Table 5.

In-distribution accuracy confirms the retrain is healthy, not just broken from less data. All three completed indist runs land within a point or two of their model’s original full-data baseline (LocalRetro: 53.30% vs. 53.11% original; MHNreact Class 4: 47.29% vs. 46.69%; MHNreact Class 8: 46.50% vs. 46.69%). Whatever happens on the held-out class is attributable to that class specifically, not to a generally degraded model.

On the held-out class itself, accuracy collapses far more completely than the frequency proxy predicted (Figure 4), **and replicates exactly across two independent classes.** The frequency proxy’s very-rare tier (freq 1–5) showed 10.0% accuracy for LocalRetro and 28.4% for MHNreact, already a severe collapse, but not remotely as severe as what genuine novelty produces: LocalRetro gets **zero out of 909 (Class 4)** and **zero out of 811 (Class 8)** correct, and MHNreact gets 1/909 and 2/811. LocalRetro’s 0% is not a one-class artifact: it reproduces exactly across two chemically unrelated held-out classes (heterocycle formation and oxidation reactions), which is what the architectural explanation below predicts and a single-class result could not have confirmed on its own. Genuine distribution shift is a strictly harder regime than the frequency proxy approximated, not just a noisier version of the same thing.

The architectural difference in vocabulary construction predicts the exact difference in outcome. LocalRetro’s 0/909 and 0/811 are not approximations of "very unlikely": they are architecturally guaranteed, since the correct templates do not exist as candidate outputs at all, and this holds identically across both held-out classes. MHNreact’s 1 and 2 correct predictions (out of 909 and 811, respectively)

Model	Class	Test	n	Accuracy	AUROC	ECE
LocalRetro	4 (Heterocycle formation)	OOD	909	0.00% (0/909)	undefined (0 correct)	0.376
LocalRetro	4 (Heterocycle formation)	Indist	4,916	53.30% (2,620/4,916)	0.726 [0.712, 0.740]	0.040
LocalRetro	8 (Oxidations)	OOD	811	0.00% (0/811)	undefined (0 correct)	0.390
LocalRetro	8 (Oxidations)	Indist	4,925	52.89% (2,605/4,925)	0.718 [0.703, 0.731]	0.044
MHNreact	4 (Heterocycle formation)	OOD	909	0.11% (1/909)	not interpretable ($n_{\text{correct}} = 1$)	0.268
MHNreact	4 (Heterocycle formation)	Indist	4,916	47.29%	0.730 [0.716, 0.744]	0.097
MHNreact	8 (Oxidations)	OOD	811	0.25% (2/811)	not interpretable ($n_{\text{correct}} = 2$)	0.257
MHNreact	8 (Oxidations)	Indist	4,925	46.50%	0.722 [0.707, 0.736]	0.095

Table 5: Held-out-reaction-class results. "OOD" = the held-out class’s reactions, entirely absent from training. "Indist" = the original test set minus the held-out class, evaluated on the same retrained model, confirming the retrain did not degrade the model generally. AUROC and ECE are computed exactly as in Sections 4–5. LocalRetro’s OOD AUROC is mathematically undefined (zero correct predictions in either class, no positive class to rank). MHNreact’s OOD AUROC is omitted for the same underlying reason it would be misleading to report: with only 1 and 2 correct predictions respectively, an AUROC value (and any confidence interval around it) reflects the rank of a single point, not a stable statistic, even though the computation runs and returns a plausible-looking number. The OOD story in this table rests on accuracy and ECE, both well-supported at these sample sizes.

are the residue of astronomically unlikely but not-forbidden guesses against randomly-initialized template embeddings. This is a clean, mechanistic explanation that falls directly out of the two models’ template-vocabulary designs, not a post-hoc story fit to the numbers, and the replication across two chemically unrelated classes for LocalRetro rules out a one-class coincidence.

Calibration is severely broken in the direction that matters most for deployment. LocalRetro’s OOD ECE is 0.376 (Class 4) and 0.390 (Class 8): the model reports substantial confidence (mean confidence roughly 38–39%) on classes where it is correct literally 0% of the time, consistently across both. MHNreact’s OOD ECE (0.268, 0.257) is similarly severe, if slightly less extreme. Both models are confidently wrong under genuine novelty, which is the practical failure mode a deployed synthesis-planning tool cannot afford. The confidence distribution underlying these ECE values is shown in Supplementary Information Section S5.

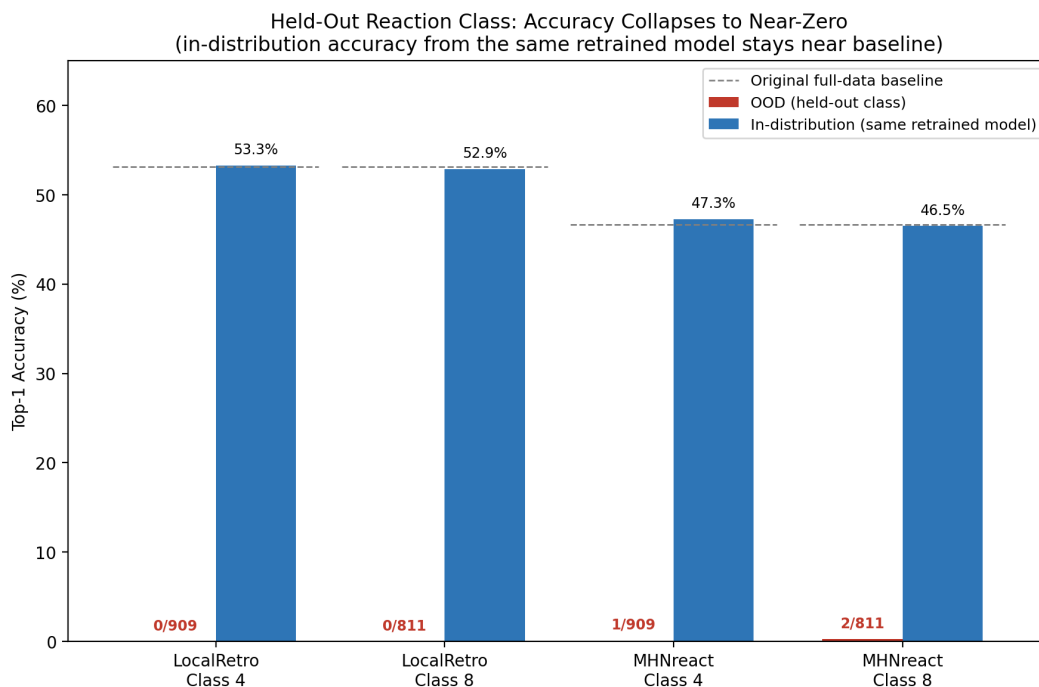


Figure 4: Held-out-class accuracy, all four model/class cells. In-distribution accuracy (blue) tracks each model’s original full-data baseline (dashed line) closely, confirming a healthy retrain. OOD accuracy (red) collapses to 0/909 and 0/811 for LocalRetro, and 1/909 and 2/811 for MHNreact, far below the frequency proxy’s very-rare-tier accuracy (10.0% LocalRetro, 28.4% MHNreact, not shown), confirming genuine distribution shift is a strictly harder regime than the proxy approximated.

7 Perturbation UQ Does Not Rescue the Rare Tail (Both Models)

The calibration collapse in Section 5 motivates an obvious question: can perturbation-based uncertainty quantification, MC dropout or a deep ensemble, recover discrimination in the rare tail where plain softmax confidence fails? The answer is no for both models, and for MHNreact, both perturbation signals do something worse than fail: they invert.

MC Dropout

For LocalRetro, no MC dropout derived signal outperformed softmax confidence. Bootstrap CIs confirm the baseline vs. MC-average comparison (0.7104 vs. 0.7063) is not a real difference: the intervals overlap completely. The very-rare tier showed an elevated MC dropout variance AUROC (0.729), but this did not survive Bonferroni correction (corrected CI [0.493, 0.901], just spanning 0.5) and was subsequently refuted by the deep ensemble below.

MC dropout was extended to MHNreact by reactivating its `nn.Dropout` submodules at inference (Section 2.3), verified against the model’s actual source rather than assumed. Per-signal AUROC for both models is shown in Supplementary Information Section S6.

MC dropout fails to improve on baseline for both models, but the character of that failure differs. For LocalRetro, all derived signals are weak-but-harmless: they simply carry less information than plain softmax confidence. For MHNreact, the failure is architecturally sharper: MC-averaging measurably *degrades* confidence (a real, statistically significant degradation, not just a null result: paired DeLong test diff = -0.0136 , $p = 7.1 \times 10^{-11}$), and dropout variance is not uninformative but actively backwards: higher disagreement across stochastic passes predicts *correct* answers more often than incorrect ones, both overall (AUROC 0.330) and in the very-rare tier (AUROC 0.361, Bonferroni CI [0.328, 0.396], entirely below 0.5).

This inversion is boundary-robust, not an artifact of one tier definition: sweeping the very-rare cutoff from $\text{freq} \leq 3$ to $\text{freq} \leq 20$, AUROC stays between 0.329 and 0.372 and the Bonferroni-corrected CI excludes 0.5 at every cut tested (Figure 5, right panel). The effect size is modest, roughly 0.13–0.17 below random, not a strong signal in the opposite direction, but it is a stable, not a fragile, finding. One benign alternative mechanism worth naming: MHNreact’s very-rare tier likely contains molecules with several near-tied candidate templates (any of which counts as correct), and dropout passes may disagree specifically on those multi-valid-template cases, which are also more likely to be scored correct regardless of which candidate is sampled. That would produce an apparent inversion without implying the model’s epistemic uncertainty is architecturally backwards; the data cannot yet distinguish this from a genuine inversion.

Deep Ensemble

To test whether epistemic uncertainty from model variance recovers discrimination in the rare-template regime where MC dropout failed, we trained 4 independent seeds of each model (LocalRetro seeds 2–5; MHNreact seeds 1–4, same architecture, hyperparameters, and verified early-stopping protocol as the baseline, differing only in initialization).

For LocalRetro, averaging predictions across the 4 seeds improves overall AUROC from 0.710 to 0.727, a real gain confirmed by a paired DeLong test and paired bootstrap (diff = 0.017, 95% CI [0.010, 0.023], $p = 3.6 \times 10^{-8}$): the one perturbation-based approach that genuinely helps, though the gain is modest. Ensemble disagreement does not recover signal in the very-rare tier (AUROC 0.601, Bonferroni CI [0.361, 0.812], spans 0.5), closing the loop on the MC dropout variance hypothesis above: disagreement across independently trained models carries no more signal in the rare tier than disagreement across stochastic dropout passes.

For MHNreact, the identical test finds no ensemble-mean improvement at all (diff = 0.001, 95% CI $[-0.006, 0.007]$, $p = 0.868$). Worse, MHNreact’s ensemble disagreement is not a clean null in the very-rare tier as it is for LocalRetro; it is inverted (AUROC = 0.427, Bonferroni CI [0.394, 0.461], entirely below 0.5), the same direction of surprise as the MC dropout variance inversion above. Very-rare-tier AUROC with Bonferroni CIs for both models is shown in Supplementary Information Section S6.

A second, completely independent 4-seed MHNreact ensemble (seeds 5–8, trained separately from the first) was run to test whether this inversion is a property of the model or an artifact of one particular set of

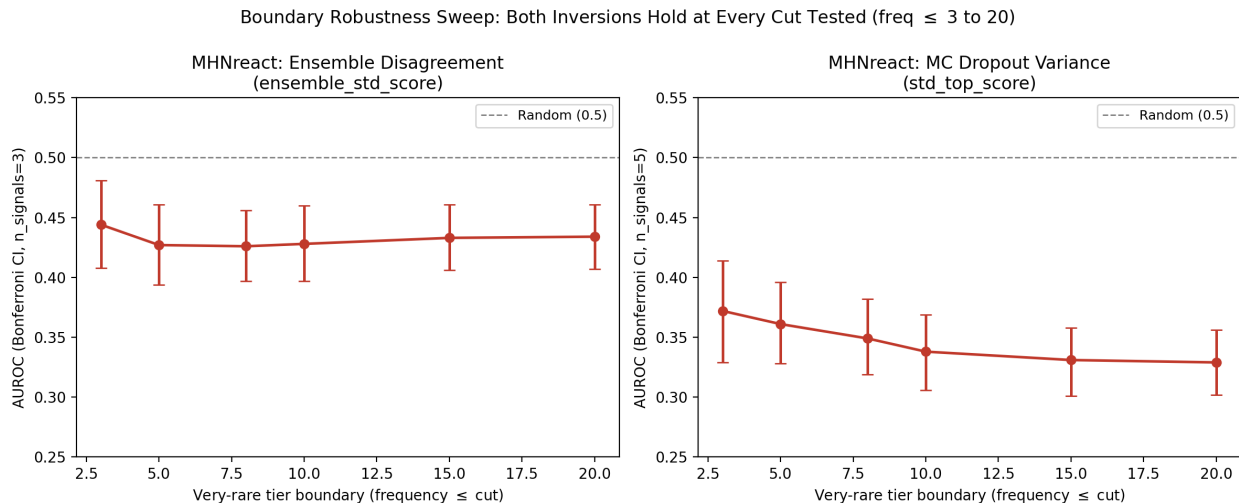


Figure 5: Boundary robustness sweep for both MHNreact inversion claims: AUROC and Bonferroni-corrected CI recomputed at six different very-rare-tier boundary definitions (freq $\leq 3, 5, 8, 10, 15, 20$). Both inversions (MC dropout variance, right; ensemble disagreement, left) hold at every cut tested: the CI excludes 0.5 in all six cases for both signals. A claim that holds only at one arbitrary boundary would be fragile; this one does not.

random initializations. It replicates closely (Supplementary Information Table S6), and a boundary sweep on this second ensemble also replicates: the inversion is not a one-ensemble or one-cutoff artifact.

Synthesis: A New Cross-Architecture Divergence

Deep ensembling was expected to behave similarly across two architecturally distinct models, mirroring the shared OOD calibration-degradation *direction* established in Section 5. It does not. LocalRetro’s ensemble mean provides a small, real overall improvement, and its ensemble disagreement is a clean null in the rare tier. MHNreact’s ensemble mean provides no improvement at all, and its ensemble disagreement is backwards in its very-rare tier: a modest effect (0.073 below random, not a strong reversal) but a statistically robust one, confirmed under a boundary sweep *and* a completely independent second ensemble.

One caveat remains regardless of how many seeds or boundaries confirm the pattern: a benign alternative mechanism is possible, structural rather than stochastic (the multi-valid-template near-ties described above), and so not something re-training or re-cutting the tier would rule out. What the evidence supports, stated at the strength it earns: an inversion that has replicated across tier-boundary choices *and* two independently trained ensembles for the deep-ensemble signal, and across every boundary tested for the MC dropout signal, arguing against assuming perturbation-based UQ fixes transfer across architectures without re-validation, though the underlying mechanism remains open. Both models still agree on the paper’s central finding, confidence calibration degrades in the rare-template tail, but the mechanisms available to detect or correct that degradation via perturbation are architecture-specific, not a shared property of template-based retrosynthesis models.

Detailed per-signal comparison tables, ROC curves, distribution plots, and the overall (non-tier) ensemble AUROC figure are in Supplementary Information Section S6.

8 What Works: Mondrian Conformal Selective Prediction (Both Models)

Section 7 showed that the obvious fixes, MC dropout and deep ensembling, do not rescue confidence in the rare tail, and can actively mislead in MHNreact. This section shows what does work: abstaining where no threshold can deliver a reliable guarantee, rather than reporting a false one.

Risk-Coverage Motivation

A single fixed target (below) is a snapshot; the risk-coverage curve shows the full tradeoff between coverage (fraction of predictions kept) and risk (error rate among kept predictions) as the confidence threshold sweeps from strictest to most permissive, and motivates why a per-tier approach is necessary at all. At a fixed 20% risk target, LocalRetro’s overall (non-tiered) curve supports keeping 22.6% of predictions; MHNreact’s supports only 14.8%, despite MHNreact having the *higher* overall baseline AUROC (0.738 vs. 0.710): AUROC is threshold-agnostic, while the risk-coverage curve also reflects the base error rate (MHNreact 53.3% vs. LocalRetro 46.9%). Per-tier, LocalRetro’s very-rare-tier curve never drops to the 20% target line even at very low coverage, visual confirmation of why Mondrian correctly abstains on that tier rather than reporting a false guarantee (full curves in Supplementary Information Section S8).

Global vs. Mondrian Thresholds

We test whether frequency-stratified (Mondrian) conformal selective prediction can restore per-tier error guarantees, using a combined softmax + MC-dropout-variance nonconformity score fit on a held-out calibration half-split, against a 20% target error rate. For LocalRetro, a global (marginal) threshold catastrophically fails: it keeps 9% of very-rare-tier predictions with a **100% error rate**. The Mondrian approach abstains on the very-rare tier almost entirely (Figure 6) (a single random split can show a sliver of coverage there, but averaged over 20 splits it is 0.8% coverage at 100% error, effectively no better than the global threshold when it does keep something), and the naive per-tier threshold lands close to, but not reliably under, the 20% target in the common and very-common tiers (20.7% and 20.2% respectively, averaged over 20 splits); the rare tier’s naive threshold overshoots the target outright (24.7%). Whether this naive accounting is trustworthy is exactly what the Hoeffding-UCB check below determines, and the answer is the same for LocalRetro as for MHNreact.

The identical harness run on MHNreact replicates the qualitative story: the global threshold badly overshoots the target in the very-rare tier (63.4% error kept vs. 20% target), and Mondrian correctly recognizes this tier is hopeless and abstains entirely. MHNreact’s global-vs-Mondrian figure and its naive per-tier results table are in Supplementary Information Section S7.

Figures 7 and 8 give the direct visual counterpart to a reliability diagram, but for the selective method itself: the target error rate as a horizontal line against the achieved error rate among kept predictions in each tier, computed from real per-molecule data and averaged over 20 random calibration/test splits rather than a single split or a schematic. For both models, the pattern is the same: the very-rare tier is effectively unusable under the naive threshold, the rare tier’s naive threshold overshoots the target, and only the common and very-common tiers come close to holding it, though even there the naive accounting is optimistic (the Hoeffding-UCB correction below shows how much).

Honesty Check: Hoeffding-UCB Correction (Both Models)

The quantitative story is not a clean win on this single calibration/test split: Mondrian’s rare tier (30.6% error kept) and common tier (22.0%) both land *above* the 20% target. Repeating the split 20 times and averaging shows the common-tier miss was single-split noise (resolves to 18.6% on average, within target), but the rare tier’s miss is real and repeats (29.0% averaged error). Neither a stricter nominal target nor dropping the inverted `std_top_score` feature fixed this; tightening the target actually collapsed coverage toward zero while the held-out error of the few remaining predictions got *worse*. This pattern indicates a sample-size/signal-strength limitation, not a mistuned parameter: the rare tier has roughly 390 calibration

Selective Prediction: Global vs. Mondrian Conformal (target error = 20%)

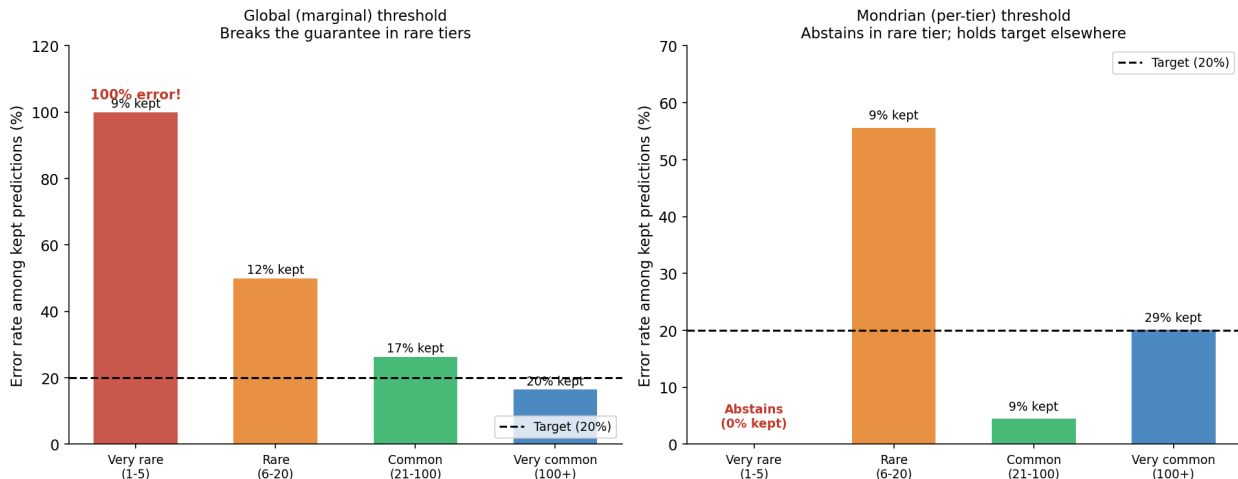


Figure 6: LocalRetro selective prediction under a 20% target error rate. *Left*: A single global threshold keeps 9% of very-rare predictions with a 100% error rate, far above the target. *Right*: Mondrian (per-tier) thresholds abstain on the very-rare tier and land close to the target in other tiers on this single split. Numbers above bars show the fraction of predictions kept (coverage); see Figure 7 for the stability-checked, 20-split-averaged version of this same result.

molecules and a 51% base error rate, close enough to chance that the logistic combiner cannot reliably find a threshold that generalizes from calibration to test.

This kind of finite-sample slack is a documented property of this exact conformal method, and we confirm directly, not just assume, that it is not a symptom specific to MHNreact: the identical Hoeffding-UCB check run on LocalRetro shows the same structural result (Table 7). Instead of selecting the most permissive threshold whose *empirical* calibration error stays under target (optimistic, since searching over many candidate thresholds and keeping the best-looking one has a selection effect), we selected the most permissive threshold whose Hoeffding *upper confidence bound* on the error rate stays under target, at 90% confidence ($\delta = 0.1$).

Tier	Naive: coverage / true error	UCB-corrected: coverage / true error
Very rare (1–5)	0.0% (abstains)	0.0% (abstains)
Rare (6–20)	8.3% / 29.0% (above target)	0.0% (abstains)
Common (21–100)	26.8% / 18.6%	0.0% (abstains)
Very common (100+)	58.7% / 19.5%	32.8% / 12.2%

Table 6: MHNreact: naive vs. Hoeffding-UCB-corrected Mondrian thresholds, averaged over 20 random calibration/test splits. The corrected threshold respects the 20% target everywhere it does not abstain, at the cost of abstaining on both the rare and common tiers entirely rather than keeping a modest fraction with an unreliable guarantee.

Both models converge on the identical structural finding, computed independently with the same procedure but different underlying signals and template vocabularies: naive per-tier thresholds are optimistic almost everywhere, and only the very-common tier, which has the most calibration data, supports a threshold that survives honest correction. This is evidence that the finite-sample optimism problem is a general property of Mondrian conformal prediction on this benchmark’s tier sizes, not an architecture-specific quirk of either model.

This makes the diagnosis precise rather than qualitative: the naive threshold was not simply “a bit optimistic” in the rare and common tiers; it was trading an invalid guarantee for coverage that a valid

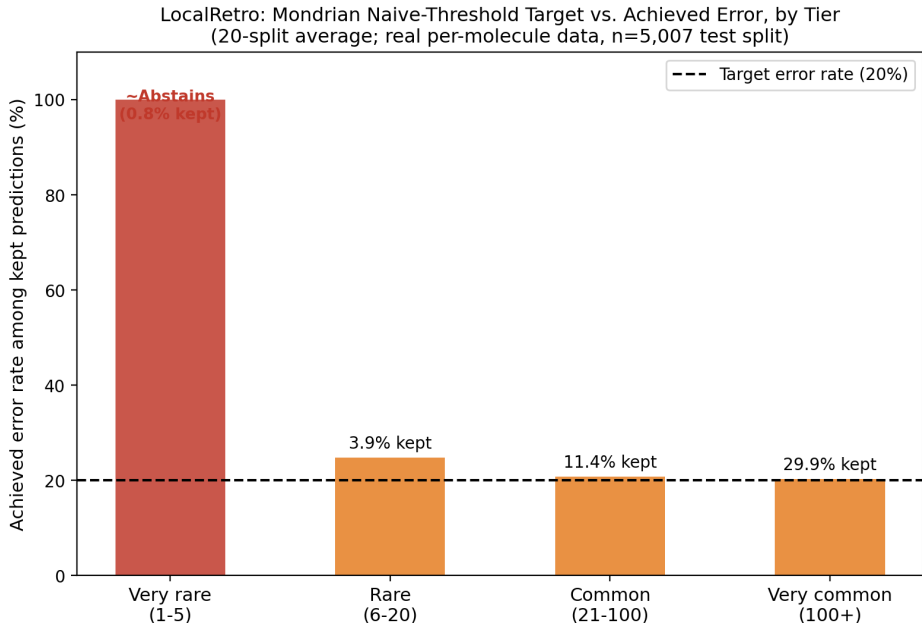


Figure 7: LocalRetro: Mondrian naive-threshold target error rate (20%, dashed line) versus achieved error rate among kept predictions, by tier, averaged over 20 random calibration/test splits from real per-molecule data ($n=5,007$ test split). The very-rare tier is marked “abstains” despite a nonzero average coverage (0.8%), since that sliver of coverage carries 100% error, no better than not predicting at all. The rare tier’s naive threshold clears the target line (24.7%); common and very-common land close to it (20.7%, 20.2%) but not reliably under it, which the Hoeffding-UCB correction addresses directly.

procedure cannot support at all with the available calibration data. Only the very-common tier (largest calibration sample, $n \approx 645$ per split) supports a real, valid selective-prediction filter.

Two further candidate fixes were tested against the same repeated-split diagnostic, and both were ruled out rather than left untested. **Calibration/test split ratio:** moving from 50/50 to 60/40, 70/30, and 80/20 (giving the rare tier’s threshold search more calibration data at the cost of a smaller held-out test set) did not help; the true held-out error *rose* monotonically with calibration fraction (29.0% $\rightarrow$ 30.2% $\rightarrow$ 31.0% $\rightarrow$ 33.9%), since shrinking the test set only adds noise to the honesty check without adding real signal to the underlying problem. **K-fold cross-conformal** ($K = 5$ and $K = 10$, where every molecule is calibration data in most folds and test data in exactly one) performed *worse*, not better: true held-out error rose to 36.3% ($K = 5$) and 43.5% ($K = 10$), and the previously-fine common tier was pushed to the edge of the target (20.1–20.3%) as well. Cross-conformal usually helps by enlarging the calibration set per fold, but here that benefit is outweighed by running the same optimistic threshold-search procedure multiple times against smaller, noisier held-out folds; with a base error rate this close to chance (51%), the repetition compounds noise rather than averaging it away.

Two further approaches were tested, differing in kind from the four above: rather than resampling the same information, they attempt to add genuinely new information or restructure the tiers themselves. **A richer nonconformity combiner** (adding `agreement_rate`, `predictive_entropy`, and `ensemble_std` to the existing `uq + std_top_score` features) barely moved the needle: true held-out error went from 29.0% to 28.3%, still far from the 20% target. **Merging the very-rare and rare tiers into one larger tier** ($n \approx 2,620$) backfired outright: rather than diluting the difficulty with more data, it abstained on every one of the 20 test splits, since the merge also blends in the very-rare tier’s 70.8% base error rate, making the target harder to hit with the combined pool, not easier.

Six independent attempts (stricter nominal target, dropping the inverted `std_top_score` feature, calibration/test split ratio, K-fold cross-conformal, a richer feature set, and tier merging) now converge on the same conclusion: this is not a fixable methodology problem within the current data. The rare tier’s

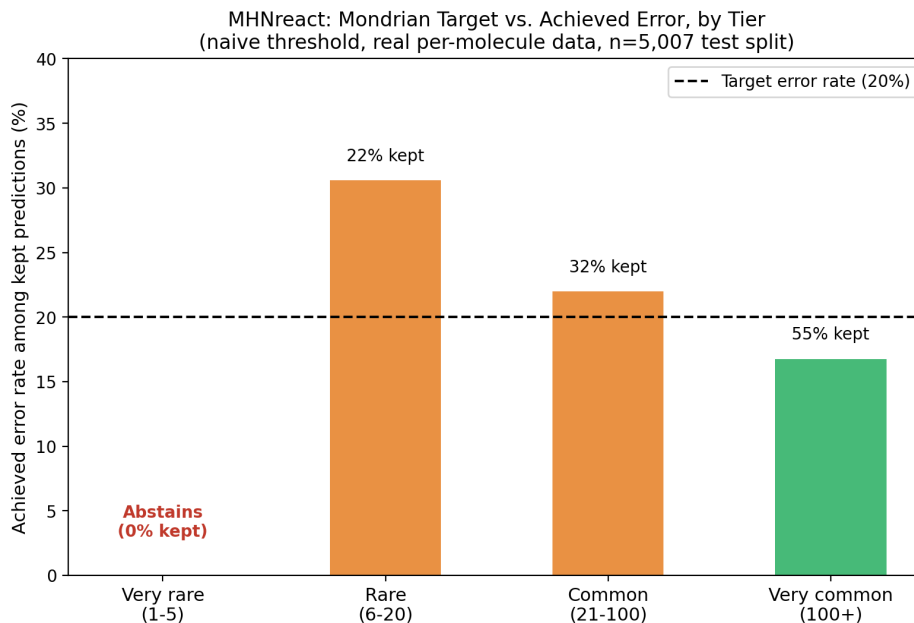


Figure 8: MHNreact: Mondrian naive-threshold target error rate (20%, dashed line) versus achieved error rate among kept predictions, by tier, computed from real per-molecule data ($n=5,007$ test split). Green bars hold the target; orange exceeds it (rare tier, 30.6%); the very-rare tier abstains entirely (0% coverage) rather than report an unreliable threshold. Coverage (fraction of predictions kept) is annotated above each bar.

discriminative signal is too weak at $n \approx 785$ total for any standard resampling or restructuring scheme to produce a threshold that reliably generalizes at a 20% target. The practical implication for deployment: the naive Mondrian threshold should not be used as-is for MHNreact’s rare or common tiers; either the Hoeffding-UCB-corrected threshold should be used as-is (honest, but close to abstaining in those tiers), or substantially more calibration data is needed, most plausibly from a larger benchmark than USPTO-50K’s fixed 5,007-reaction test set, since no resampling or restructuring of the existing pool closes the gap.

Tier	Naive: coverage / true error	UCB-corrected: coverage / true error
Very rare (1–5)	0.8% / 100.0% (far above target)	0.0% (abstains)
Rare (6–20)	3.9% / 24.7% (above target)	0.0% (abstains)
Common (21–100)	11.4% / 20.7% (marginally above target)	0.0% (abstains)
Very common (100+)	29.9% / 20.2%	10.0% / 14.2%

Table 7: LocalRetro: naive vs. Hoeffding-UCB-corrected Mondrian thresholds, averaged over 20 random calibration/test splits, computed with the same procedure as Table 6. The structural result matches MHNreact exactly: the naive threshold is optimistic in every tier except very-common, and only the very-common tier supports a valid selective filter once the optimism is corrected for, at reduced coverage (10.0% vs. MHNreact’s 32.8%) but a lower, more conservative achieved error (14.2% vs. 12.2%, both comfortably under the 20% target).

9 Discussion

Several results (the perturbation-UQ negative result, the calibration collapse shared across two architectures, and the divergent perturbation-UQ behavior discovered along the way) combine into a coherent picture of template-based retrosynthesis confidence.

- Softmax confidence is a real discriminator overall in both models: LocalRetro AUROC 0.710 [0.696, 0.724], MHNreact AUROC 0.738 [0.724, 0.752]. Both are moderate but statistically robust signals.
- Both models show degradation concentrated in the rare template tail. Using the rank-based vocabulary-matched tiers (Table 2), the debiased-ECE degradation ratio (rare/high) is $3.2\times$ for LocalRetro and $2.9\times$ for MHNreact (similar by that denominator), but the rare/mid ratio ($7.3\times$ vs. $1.8\times$) shows the underlying failure shapes differ substantially. The shared degradation *direction*, not a single ratio, is the cross-architecture finding.
- The two models exhibit different failure signatures within that shared trend (Figure 2): LocalRetro is well-calibrated in-distribution and collapses under novelty; MHNreact is mildly but uniformly miscalibrated across all tiers, with no sharp collapse. These are two architecturally distinct miscalibration regimes sharing one qualitative property: worst calibration in the rare tail.
- Retiering by predicted rather than true template frequency (Section 5), the deployment-realistic scenario, confirms this same architectural distinction shows up even in the metric used to define the tiers themselves. Wrong predictions default to common templates in both models, which partially inverts LocalRetro’s sharp gradient and compresses MHNreact’s diffuse one, without changing which model has which kind of failure.
- MC dropout adds no discrimination over plain softmax confidence overall for LocalRetro, and the $20\times$ inference cost is not justified for in-distribution error detection. For MHNreact, MC dropout is not merely useless but actively counterproductive: MC-averaging degrades confidence, and dropout variance is inverted.
- A 4-seed deep ensemble provides a real overall improvement for LocalRetro ($+0.017$ AUROC, $p = 3.6 \times 10^{-8}$), the one perturbation-based approach that genuinely helps, though the gain is modest. This does not generalize to MHNreact: no ensemble-mean improvement, and an ensemble-disagreement signal that is inverted rather than merely null in the very-rare tier, replicated across boundaries and a second independent ensemble. Deep ensembling is not a portable UQ fix across architectures, even though both models share the same underlying calibration-degradation direction.
- Mondrian (per-tier) conformal selective prediction is the practical, architecture-agnostic fix. A global threshold catastrophically over-keeps very-rare predictions (100% error rate on the 9% it keeps for LocalRetro). Per-tier thresholds abstain almost entirely on the very-rare tier and land close to the error target elsewhere for both models on a single calibration split, though the Hoeffding-UCB check, run independently on both architectures, shows the naive per-tier guarantee is itself optimistic in every tier except very-common: only the largest-calibration-sample tier supports a threshold that survives honest correction in either model. This is a genuine, non-trivial limitation of the naive method at this benchmark’s tier sizes, not a second-order caveat specific to one architecture.

Template-based retrosynthesis models are honest about what they know. That is probably why MC dropout and deep ensemble disagreement, in-distribution, find little room to add beyond softmax confidence in LocalRetro. The OOD calibration degradation is the most consequential result for deployment: confidence becomes poorly calibrated exactly on the rare, novel templates where a synthesis planning tool most needs a reliable “I don’t know” signal. This degradation is shared across two architecturally distinct models with matched ECE ratios of $\sim 3\times$, suggesting it reflects the task’s distributional structure (a template vocabulary with a long-tail frequency distribution creates a hard coverage boundary) rather than any specific modeling choice. The Mondrian conformal fix operationalizes this diagnosis into a practical decision rule with per-tier guarantees applicable to any model in this family. Deep ensembling and MC dropout, however, do not share that portability: they help LocalRetro modestly (ensemble) or not at all (dropout), and actively mislead for

MHNreact, so any deployment relying on perturbation-based disagreement as a rare-tier safety signal needs architecture-specific validation rather than an assumption of transfer.

10 Limitations and Next Steps

- All results are on USPTO-50K. Whether the calibration collapse, the perturbation-UQ inversions, and the Mondrian mitigation generalize to a larger, longer-tailed benchmark (USPTO-full, USPTO-MIT) is untested; the failure modes reported here could be more or less severe at scale.
- The held-out-class experiment covers two reaction classes (heterocycle formation, oxidations), selected for template exclusivity rather than chemical breadth. Replication across two chemically unrelated classes for LocalRetro is reassuring but does not establish that every novel reaction type would collapse identically.
- The main-text calibration analysis (Section 5) primarily tiers by template training frequency as a proxy for distributional novelty; this proxy correlates with but is not identical to genuine distribution shift. Section 5 directly tests the deployment-realistic alternative, tiering by the model’s own predicted template rather than the (inference-time-unknown) true template, and finds the picture changes for both models: LocalRetro’s gradient partially inverts and MHNreact’s compresses, though neither model’s qualitative failure signature changes. Extending this predicted-tiering analysis to the Mondrian conformal thresholds themselves, rather than the descriptive accuracy tiering reported here, remains future work.
- LocalRetro’s original baseline checkpoint could not be recovered for the predicted-template-tiering analysis (Section 5): LocalRetro’s training code sets no explicit random seed, so retraining does not reproduce a lost checkpoint any more precisely than substituting an existing one. The analysis therefore uses ensemble checkpoint seed2 in place of the original, with 20.9% of predictions excluded as landing in a reserved “no resolvable template” output class. Every other LocalRetro result in this paper uses the original baseline checkpoint.
- Both models’ Mondrian calibration pools are too small in the rare and common tiers (MHNreact $n \approx 785$; LocalRetro’s rare and common tiers are smaller still) for the naive per-tier threshold to generalize reliably; the Hoeffding-UCB-corrected threshold is honest but abstains on most tiers as a result, confirmed independently for both architectures (Section 8). This is a sample-size limitation of the current benchmark, not of the method.
- The mechanism behind MHNreact’s inverted perturbation-based UQ signals (MC dropout variance, ensemble disagreement) is not identified. A benign explanation (near-tied multi-valid-template cases) and a genuine architectural cause remain equally consistent with the data; distinguishing them would require molecule-level inspection rather than further statistical testing.
- The Mondrian nonconformity score is a simple logistic combination of two features (softmax confidence and MC-dropout variance). More expressive combiners were tested only within the current calibration pool’s sample-size constraints and produced negligible improvement; whether a richer score would help given more data is untested.
- Each model’s headline results rest on a single trained checkpoint (with independent ensembles used only for the ensemble-specific analyses). Run-to-run variance beyond what those ensembles capture is not separately quantified.
- Temperature scaling (Section 5) uses a one-vs-rest approximation on the top-1 probability, since the full softmax distribution over the template vocabulary was not retained; results may differ modestly from temperature scaling applied to the full logit vector.

Data and Software Availability

All analysis code (calibration toolkit, AUROC bootstrap, Mondrian conformal harness, quantile binning, paired DeLong testing, temperature scaling, and boundary sweep implementations) is available on Github:

<https://github.com/yamenfargaly-maker/Uncertainty-Quantification-with-Template-Based-Retro-synthesis-Models>. Trained model checkpoints are available at the same repository: LocalRetro’s 4-seed ensemble (seeds 2–5) and both held-out-class checkpoints, and MHNreact’s baseline and two independent 4-seed ensembles. LocalRetro’s original baseline checkpoint could not be included, as it was lost prior to this release and is not recoverable (see Limitations). USPTO-50K, the dataset underlying every result in this paper, is publicly available from its original source (Schneider et al. 2016 reaction role classification; Lowe 2012 patent reaction extraction).

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